

Assignments

Topic-wise practice worksheets, one folder per week of the plan in `../docs/03-practice-plan.md`. Each folder (once built) contains:

- `assignment.md` — the questions (4-option MCQ, Class 4 difficulty, mixing direct computation with word-problem/Everyday-Maths style framing)
- `answer-key.md` — correct options with a short worked explanation each

Folder list (per the 12-week plan)

Folder	Topic	Status
<code>01-number-sense/</code>	Place value, number names, ordering, rounding, Roman numerals	✓ Built
<code>02-four-operations/</code>	Addition, subtraction, multiplication, division (multi-digit + word problems)	✓ Built
<code>03-factors-multiples-patterns/</code>	Factors, multiples, number patterns	✓ Built
<code>04-fractions-decimals/</code>	Comparing & equivalent fractions, simple decimals	✓ Built
<code>05-measurement-money/</code>	Length, weight, capacity, money, conversions	✓ Built
<code>06-time-calendar/</code>	Clock time, duration, calendar	✓ Built
<code>07-geometry-perimeter-area/</code>	Shapes, symmetry, perimeter, basic area	✓ Built
<code>08-data-handling/</code>	Tally marks, pictographs, bar graphs	✓ Built
<code>09-reasoning-patterns-coding/</code>	Number/figure patterns, analogy, classification, alphabet test, coding-decoding	✓ Built
<code>10-reasoning-direction-figures/</code>	Ranking, direction sense, days & dates, mirror images, embedded figures, geometrical solids	✓ Built (24 Q)
<code>11-achievers-hots/</code>	Multi-step HOTS problems across all topics above	✓ Built
<code>12-math-kangaroo-stretch/</code>	Optional — Math Kangaroo-style puzzles (shape counting, mirror images, logic, cube nets)	✓ Built (15 Q, 3 tiers)

Folders `01 – 11` are the core 12-week plan, built in the same two-file (`assignment.md` + `answer-key.md`) shape, 15 questions each — except `10-reasoning-direction-figures/`, which has 24 (15 text-based + 9 diagram-based). `12-math-kangaroo-stretch/` is the optional extension

described in `docs/03-practice-plan.md`, not part of the core plan or any Indian olympiad's syllabus. `08-data-handling/`, `10-reasoning-direction-figures/`, and `12-math-kangaroo-stretch/` each ship an `images/` folder of SVG diagrams referenced from their `assignment.md`.

Diagram-based question types

Mirror images, embedded figures, and geometrical-solid recognition are inherently visual Section-1 (Logical Reasoning) sub-topics. Rather than describing them in text, `10-reasoning-direction-figures/images/` has an SVG diagram for each question:

- **Mirror images** — the original figure plus 4 candidate reflections (one correctly flipped left-right, the other three are unchanged / flipped upside-down / rotated 180° — classic distractors). The correct option is generated by an exact `scale(-1,1)` transform, not hand-drawn, so it's guaranteed accurate.
- **Embedded figures** — a composite figure made of two overlapping shapes; find which of 4 candidate shapes is traceable inside it.
- **Geometrical solids** — a drawn cube, cone, and cylinder, each with a question about its name or face count.

`08-data-handling/images/` similarly has a real SVG pictograph and bar graph (rather than an emoji/markdown-table stand-in) for its data-handling questions.

`12-math-kangaroo-stretch/images/` has 7 more SVGs for its puzzle questions — shape-counting figures, a mirror-image puzzle (same guaranteed-correct transform technique as above), a balance-scale illustration, a grid for path-counting, a shaded-fraction figure, and a labelled cube net.